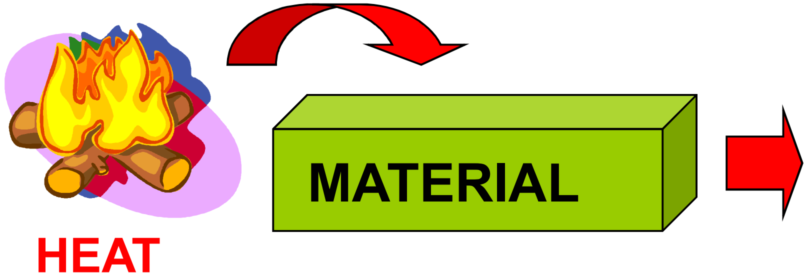
THERMAL PROPERTIES OF MATERIALS

THERMAL PROPERTIES: knowledge of materials response to the application of heat.



- Temperature rise
- Dimensions increase
- Energy transport to cooler regions
- Melting

RELEVANT THERMAL PROPERTIES:

- Heat capacity
- Thermal conductivity
- Thermal expansion
- Thermal stresses

The knowledge of **materials behaviour at high temperatures** and **how their properties change with temperature** is fundamental for material applications

Mechanisms of heat transfer in a solid material

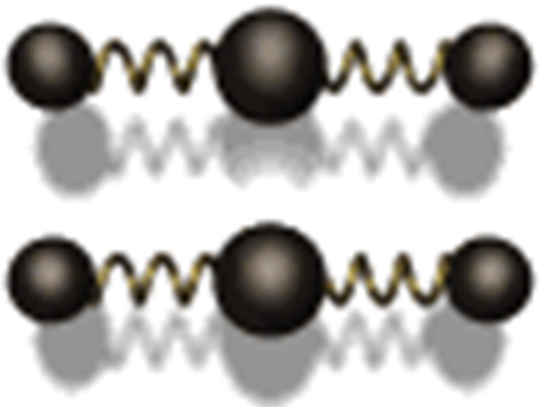
Thermal energy is transferred:

- both by free electrons (for instance in metals, where there is an "electronic cloud")
- and by vibrational waves (for instance in ceramics, where there are no free electrons).

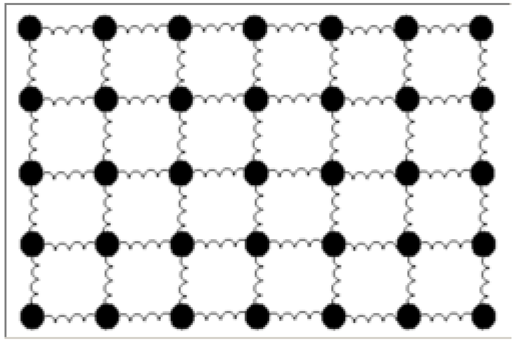
If heat is supplied to a solid material:

→ atomic vibrational energy increases

→ atoms are not independent (due to their chemical bonds): heat is transferred through the material (crystalline lattice or amorphous structure) by vibrational waves called **PHONONS**



Considering the regular lattice of atoms in a uniform solid material



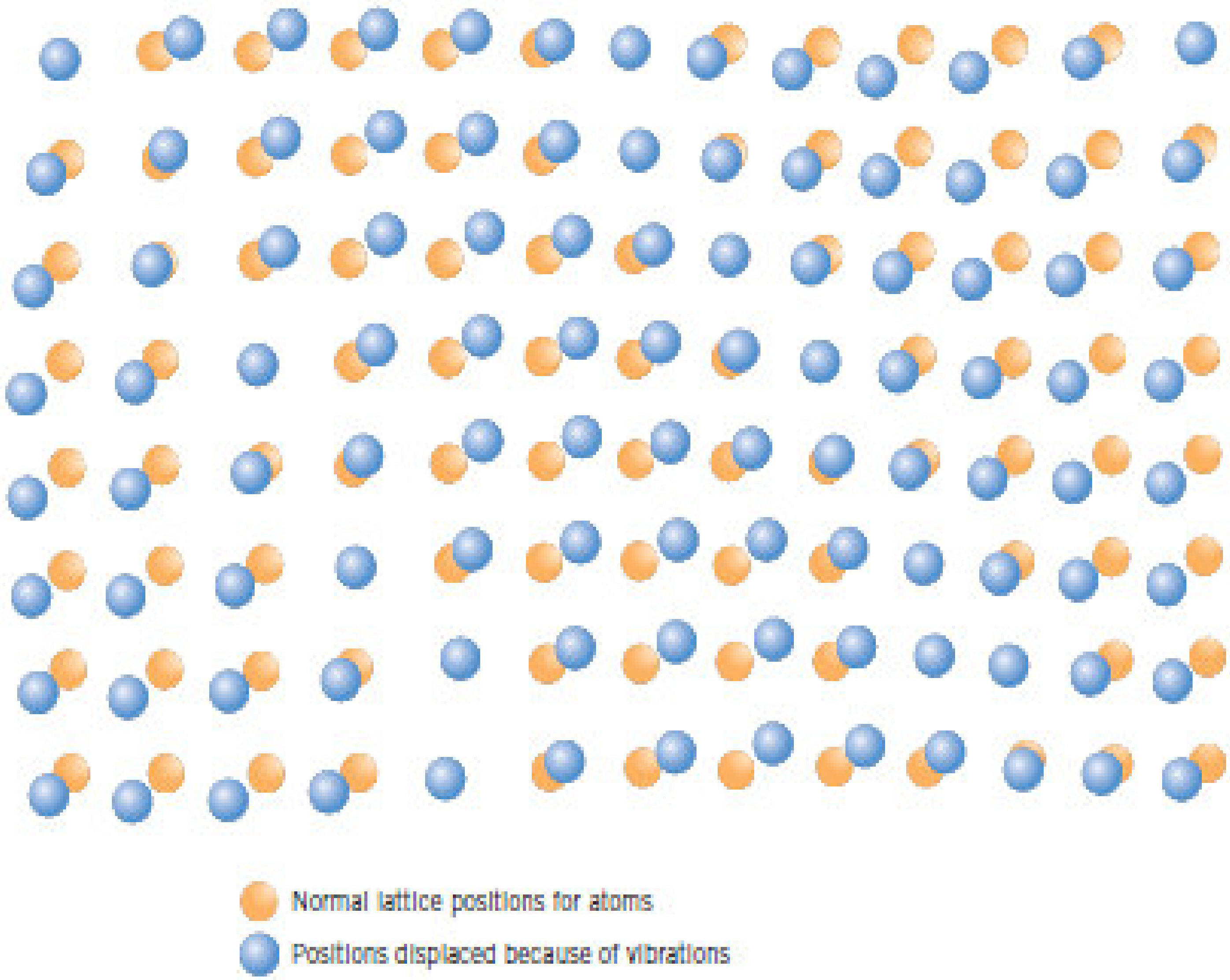
Atoms are tied together with bonds, so they can't vibrate independently. The vibrations take the form of collective modes which propagate through the material.

Propagating lattice vibrations can be considered to be sound waves, and their propagation speed is the speed of sound in the material.

The modes of vibration involve the entire crystal volume.

In real crystals, amplitude of vibration is typically much smaller than the lattice spacing.

Figure 19.1 Schematic representation of the generation of lattice waves in a crystal by means of atomic vibrations. (Adapted from "The Thermal Properties of Materials" by J. Ziman. Copyright © 1967 by Scientific American, Inc. All rights reserved.)



https://www.youtube.com/watch?v=RgRwHL4P8tU

The internal energy of a solid material increases as heat is provided to it.

Thermal vibration amplitude increases as heat is supplied to the material

↓

temperature increases → the **kinetic energy of electrons** increases and the **interatomic distance** increases → **the solid changes its size**

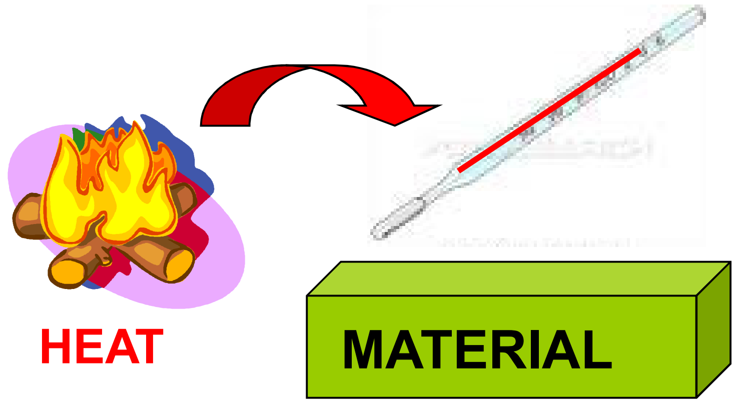
Thermal vibrations:

- **Atomic, ionic and molecular** vibrations around the equilibrium positions of an atom/ion/molecule in the lattice
- **Atomic** vibration **inside molecules**
- **Rotation** of molecules

During **phase transformations** (e.g. transitions of state of matter) heat is spent to do the transformation and **there is no temperature increase**

Heat capacity

This parameter denotes the material ability to **absorb heat** from the external surroundings.



The heat capacity, **C** (J/K), of a system is the ratio between the heat added to a material (dQ) and the resultant change in its temperature (dT):

$$C = \frac{dQ}{dT}$$

Practically, it is the ratio between the absorbed thermal energy ΔE and the resultant increase in temperature ΔT:

$$C = \frac{\Delta E}{\Delta T}$$

To make a comparison of heat capacities of different materials, we introduce a new parameter derived from **C**:

the ***SPECIFIC HEAT*** (c) represents the heat capacity per unit mass (i.e. 1 kg) to raise a material's temperature by 1 degree (1 K)

$$c = \frac{dQ}{m \cdot dT} \qquad \text{(J/kg K)}$$

We can define 2 types of specific heat measured according to different conditions:

→ c_p: measured with constant external pressure

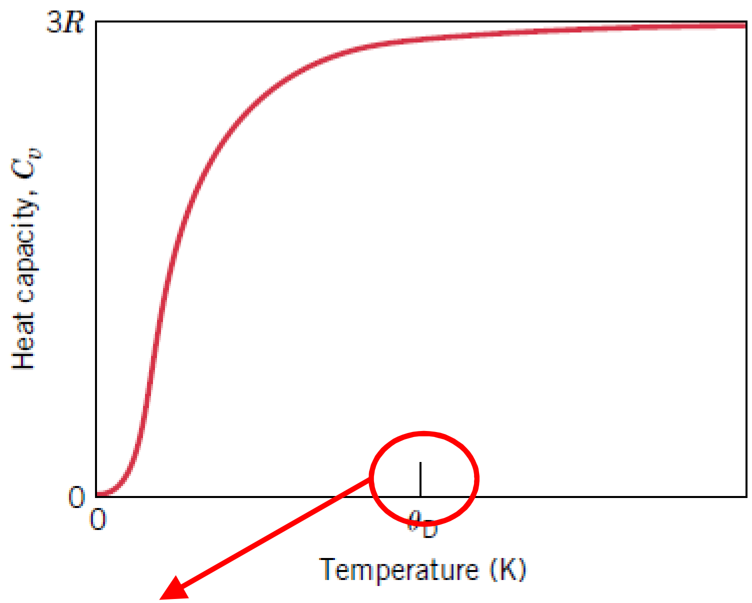
→ c_v: maintaining specimen volume constant

$c_p > c_v$ but the difference is very small at room temperature, so we can usually define a single value of specific heat **c**.

The heat capacity **C** (and accordingly the specific heat **c**) **increases as T increases** (at relatively low temperature):

→ it is necessary to give a bigger amount of heat to raise temperature of a former heated material

→ above a certain temperature it becomes constant



DEBYE TEMPERATURE
(below room temperature for many solid materials)

If the specific heat, **c**, is referred to 1 mole of substance (instead of 1 kg we consider a number of atoms or molecules equal to the Avogadro's number, $N_A = 6\times10^{32}$), it is called **ATOMIC OR MOLECULAR HEAT** (J/mole K). This parameter is preferably used in chemistry and materials science.

For most of solid elements the ***ATOMIC HEAT is about 25 J/mol K***.

→ atomic heat depends on temperature at low T

→ at high T it is approximately constant (or slightly over 25 J/mol K)

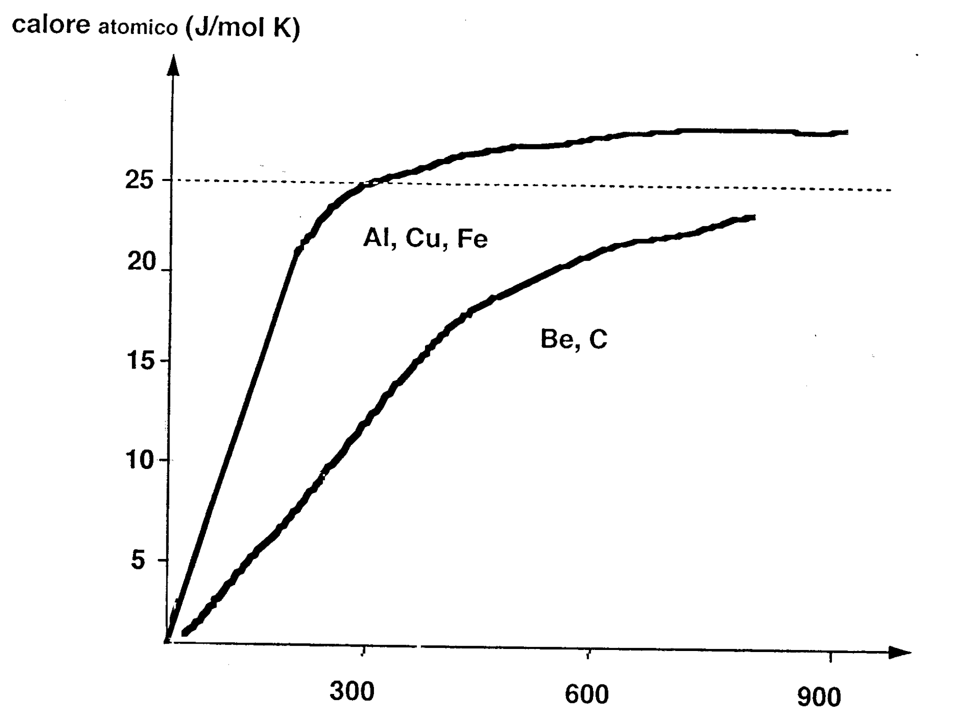
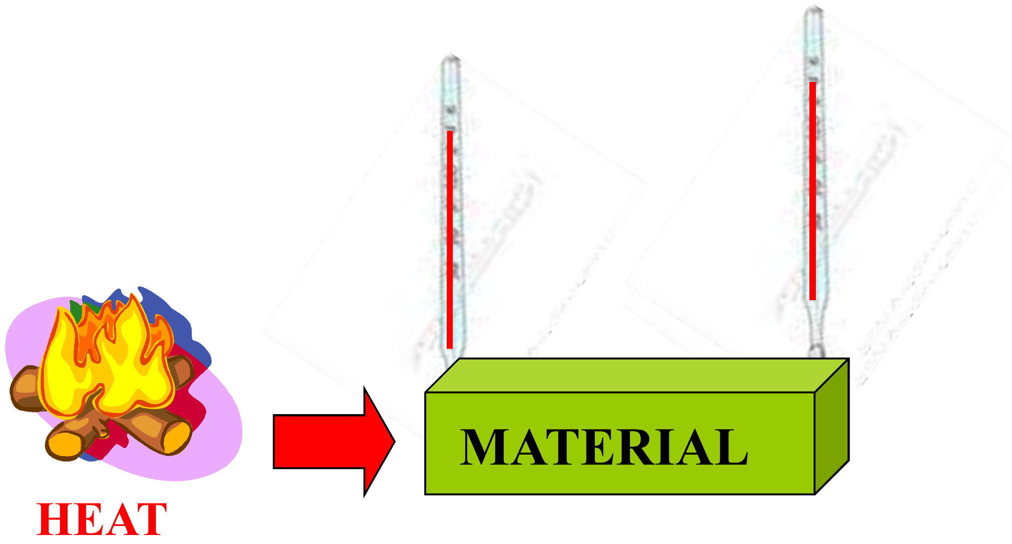


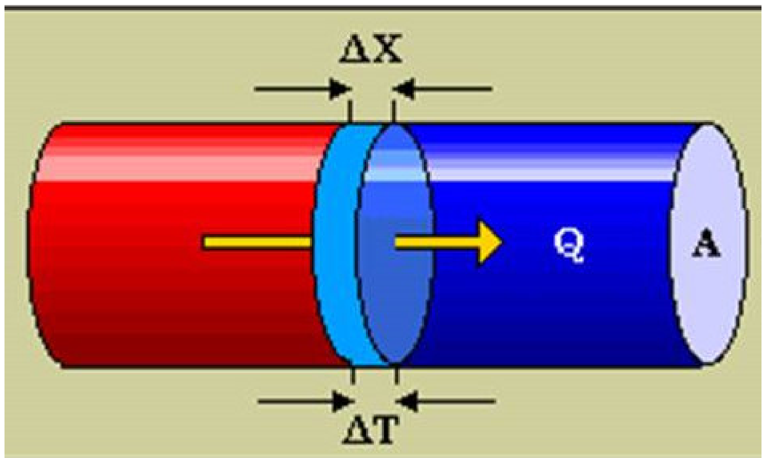
Figura 1

Heat capacity and derived parameters depend on the atomic weight.

Thermal conductivity

Thermal conductivity is the **ability of a material to transfer thermal energy** from a system at high temperature to another system at lower temperature.





We define the thermal power Q (W) as:

$$Q = \boxed{k} \cdot A \cdot \frac{\Delta T}{\Delta x}$$

1-D situation, e.g. in a wire

THERMAL CONDUCTIVITY COEFFICIENT

$$q = \frac{Q}{A} = k\frac{dT}{dx} \qquad \frac{\mathrm{W}}{\mathrm{m}^2}$$

Thermal flow

Example

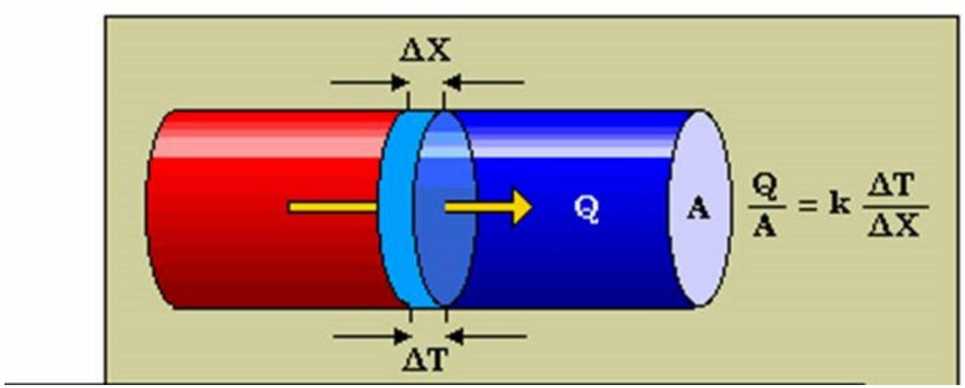
Calculate the heat flowed in 1 hour through a 2 mm thick polymeric sheet (1 m²) when one side is at 20 °C and the other side is at 50 °C.

$k = 0.58$ W/m K

1 W=1 J / s

$Q = 1 \times 0.58 \times 30 / (2\times 10^{-3}) = 8700$ W

Heat in 1 h = $8700 \times 3600 = 3.1\ 10^{7}$ J



Mechanisms of heat conduction

- Lattice vibration waves (PHONONS)
- Free electrons

TOTAL CONDUCTIVITY

$$k = k_l + k_e$$

LATTICE VIBRATION CONDUCTIVITY

Phonon net movement

ELECTRON CONDUCTIVITY

Kinetic energy transfer

First mechanism (**free electrons**) is predominant in **metals**: it is related to electrical conductivity (also in this case we have movement of electrons) and it is very efficient → metals have **high thermal conductivity**. It can also depend on impurities and alloying elements.

Second mechanism (**phonons**) is predominant in **ceramics** and it is less efficient → in fact ceramics are **insulating materials**

Thermal conductivity of **glasses** is **lower than thermal conductivity of ceramics** → vibrational waves very hardly propagate without crystalline lattice

Porous materials → **low thermal conductivity** (air entrapped inside the pores is a bad heat conductor → desired thermal insultating effect)

Thermal conductivity of **polymers** is **lower than thermal conductivity of ceramics** (heat is partially dissipated by vibration/rotation of polymeric chain molecules); it is as much lower if polymers are amorphous

In general, thermal conductivity depends on temperature.

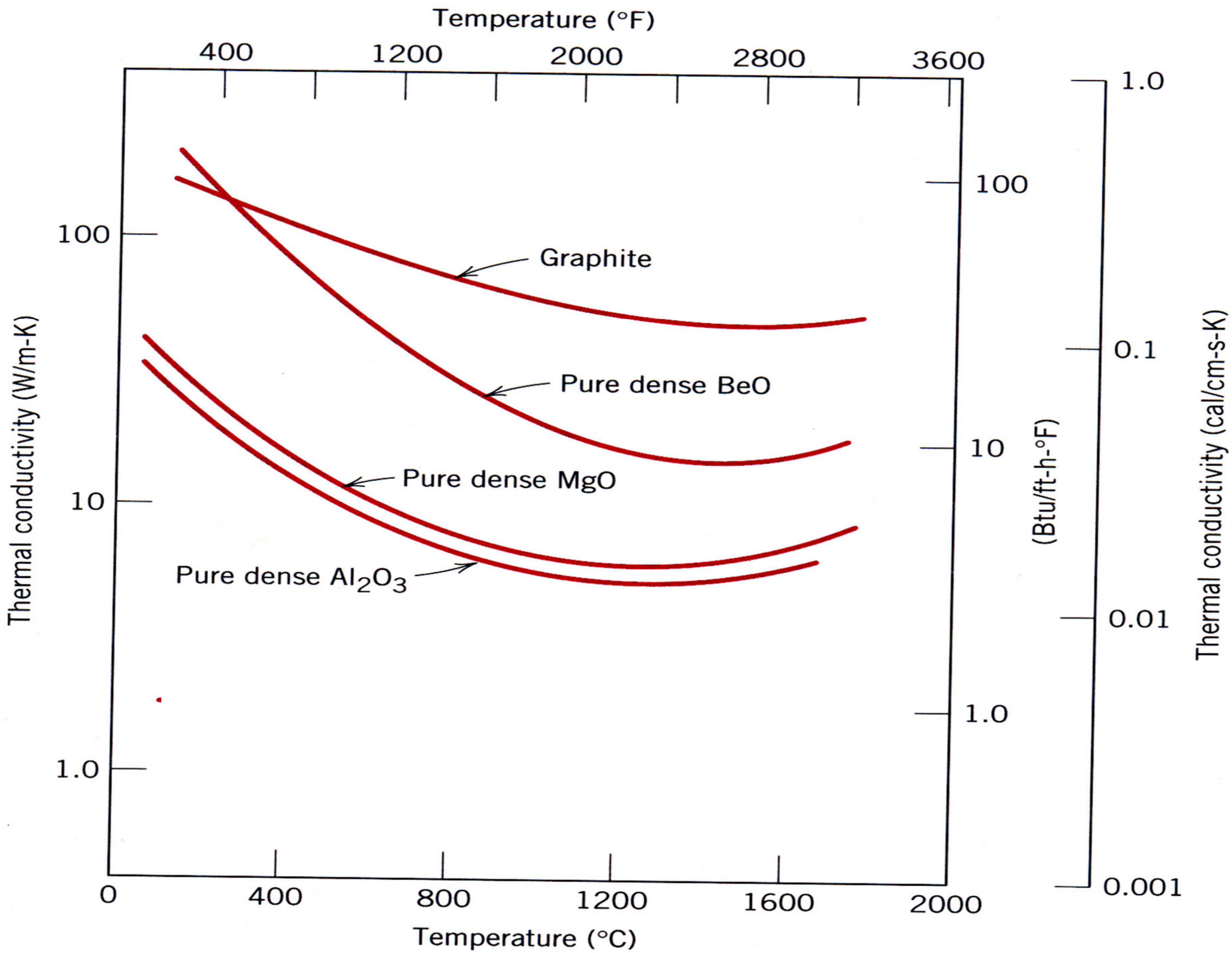
Random thermal agitation increases with increasing temperature → disturbing effect on propagation of phonons

(for instance **k_l decreases with the increase of T in the case of ceramics**).

Electrons motion does not change (significantly) with T

(practically, **k_e does not change with T in the case of metals**)

However, remember that **BOTH MECHANISMS** are always present in solid materials!



Temperature (°F)
400
1200
2000
2800
3600
Thermal conductivity (W/m-K)
100
10
1.0
(Btu/ft-h-°F)
100
10
1.0
Thermal conductivity (cal/cm-s-K)
1.0
0.1
0.01
0.001
Graphite
Pure dense BeO
Pure dense MgO
Pure dense Al_2O_3
0
400
800
1200
1600
2000
Temperature (°C)

Material	k (W/m-K)[c]
Aluminum	247
Copper	398
Gold	315
Iron	80.4
Nickel	89.9
Silver	428
Tungsten	178
1025 Steel	51.9
316 Stainless steel	16.3[d]
Brass (70Cu–30Zn)	120
Alumina (Al_2O_3)	30.1
Beryllia (BeO)	220[e]
Magnesia (MgO)	37.7[e]
Spinel ($MgAl_2O_4$)	15.0[e]
Fused silica (SiO_2)	2.0[e]
Soda–lime glass	1.7[e]

METALS

CERAMICS

In most cases, metals exhibit <u>higher thermal conductivity</u> than ceramics.

$$k_{metals} > k_{ceramics} >> k_{polymers}$$

Some practical effects of heat capacity and thermal conductivity

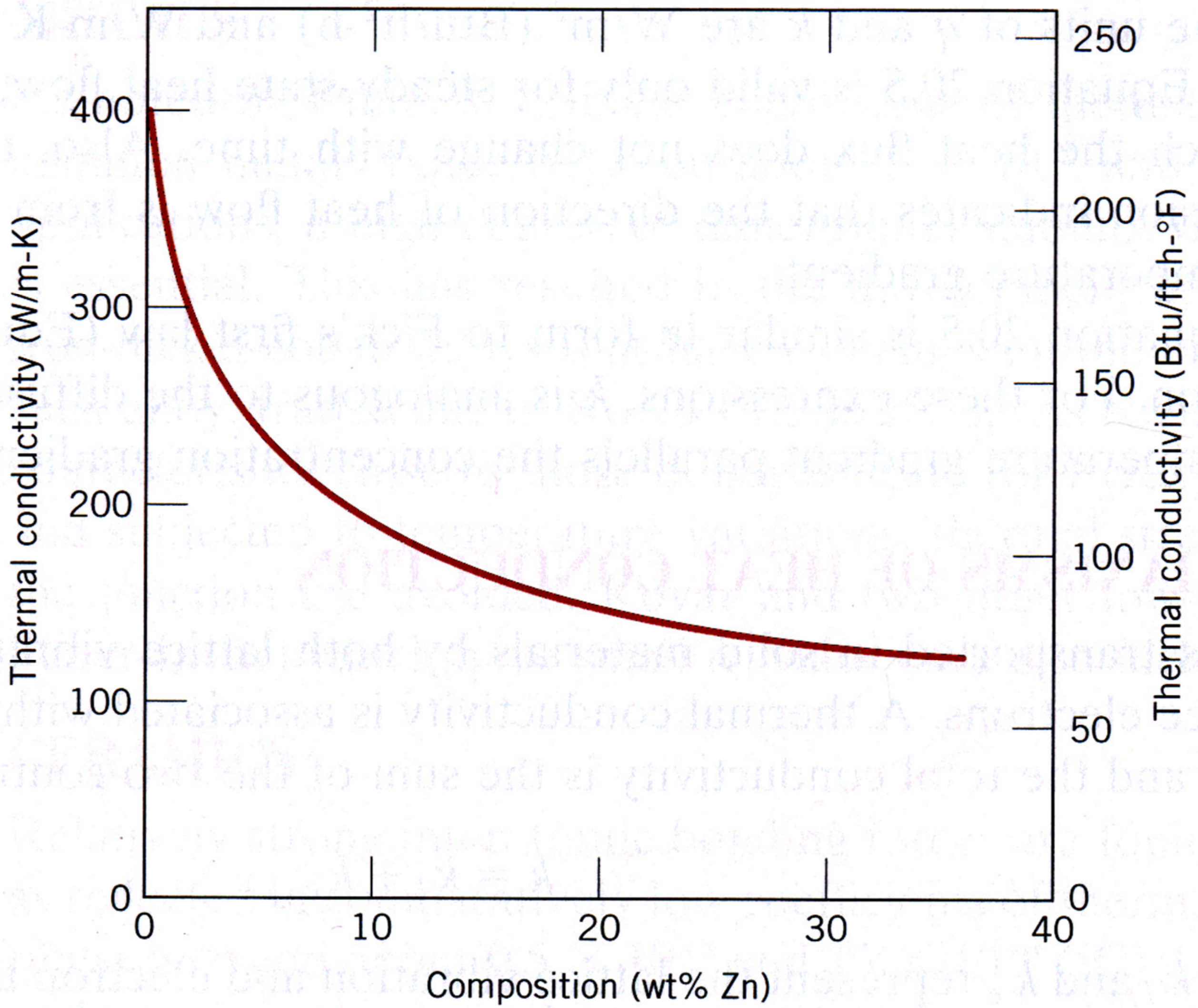
A component made by using a material with high specific heat takes more time (in other words, more heat) to increase its temperature in comparison with a component made of a material showing a low specific heat.

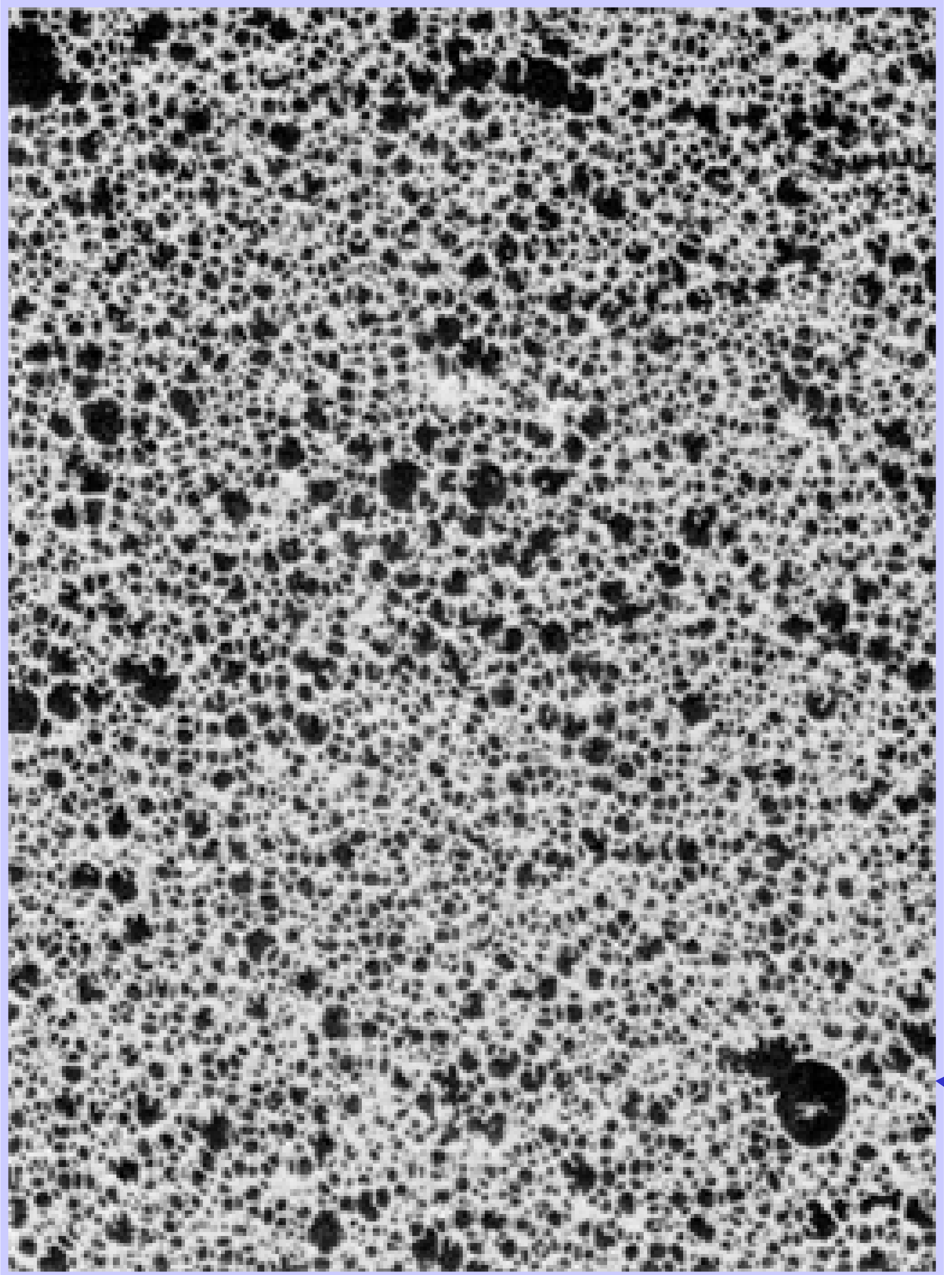
Example:

Why is steel cold to touch?

Because steel has high thermal conductivity and high specific heat; as a consequence, it is cold to the touch → heat flux from our body is significant (high thermal conductivity) and a big amount of heat is needed to raise steel's temperature from room temperature (25 °C) up to equilibrium with our hand (37 °C) (steel has high heat capacity).

k vs Zn % in Cu-Zn alloys





POLYMERS

$k \sim 0.3$ W/m K

Vibration and rotation of molecular chains

k increases with the % of crystalline structure

Typical insulating materials

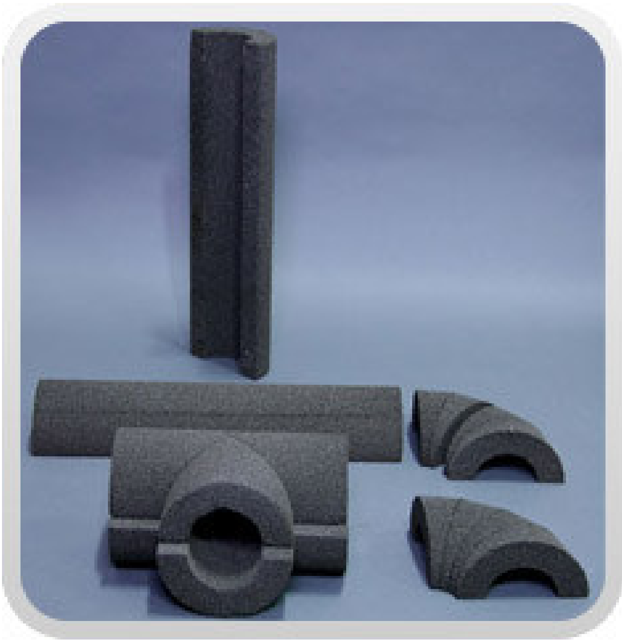
Polymeric Foams: role of pores

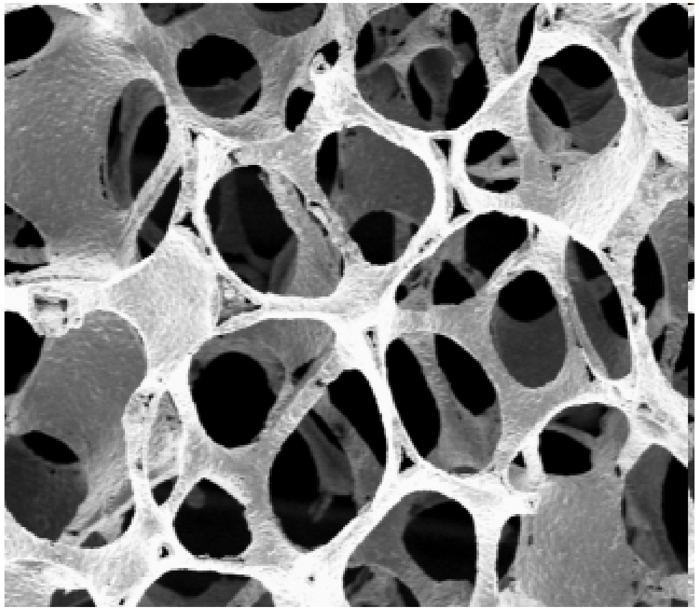
Role of porosity on thermal conductivity

- k (air) = 0.02 W/m K

Foam materials

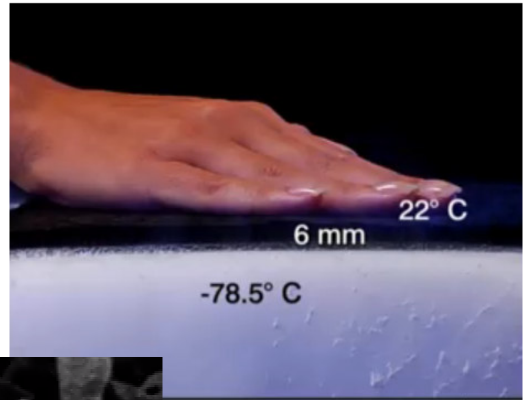
Foam glass, k= 0.04 W/mK

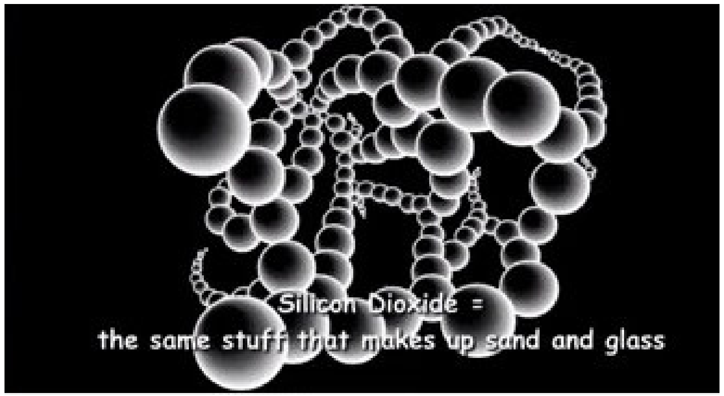


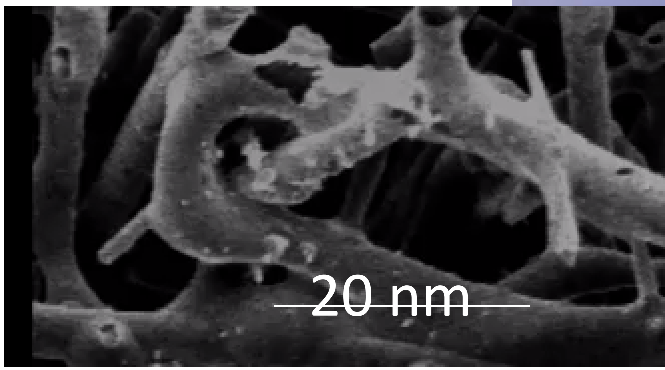


Silica aerogel

- One of the lowest density material (1.6-1.9 mg/cm^3)
- 95% air, 5% SiO_2
- pore size about 20 nm

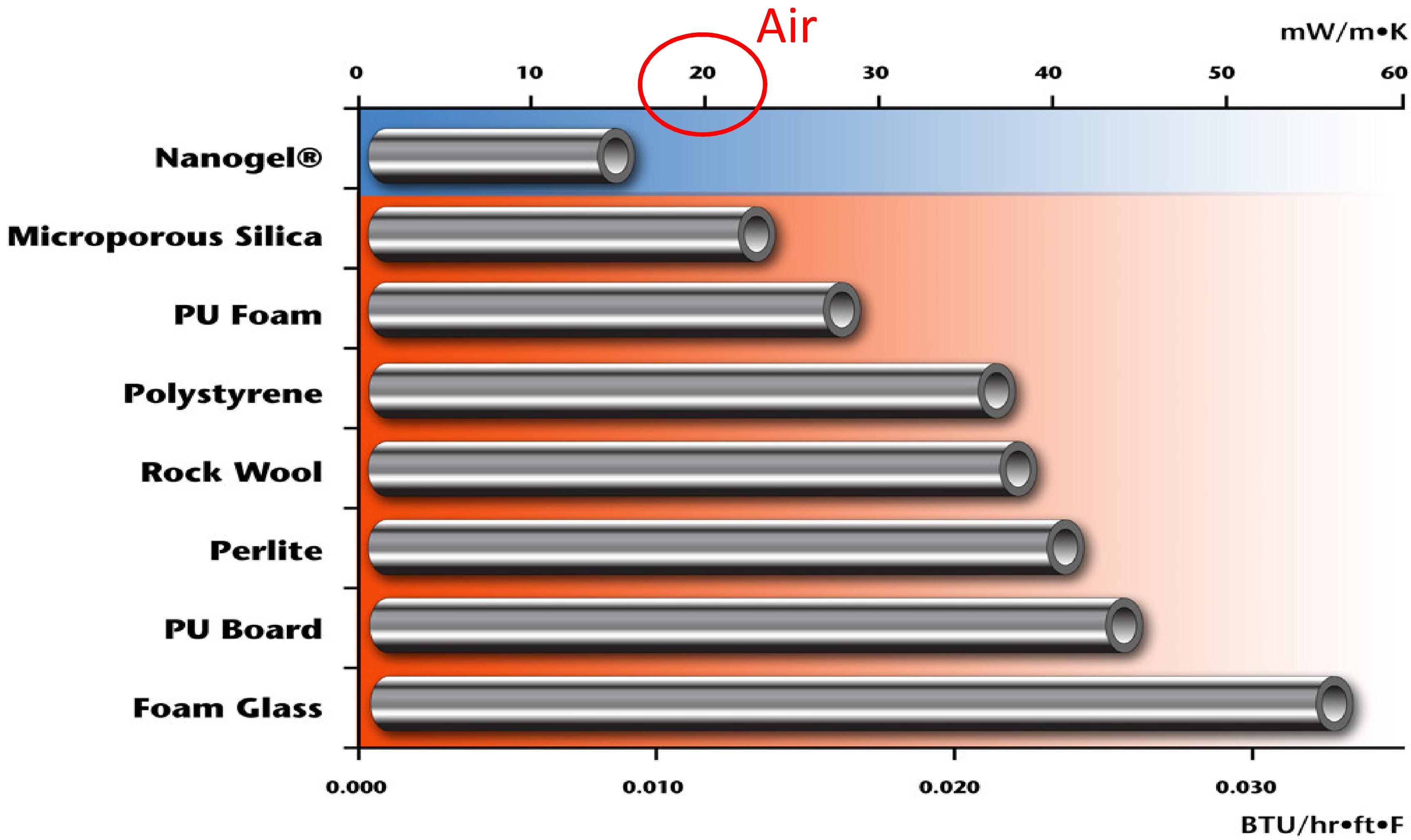




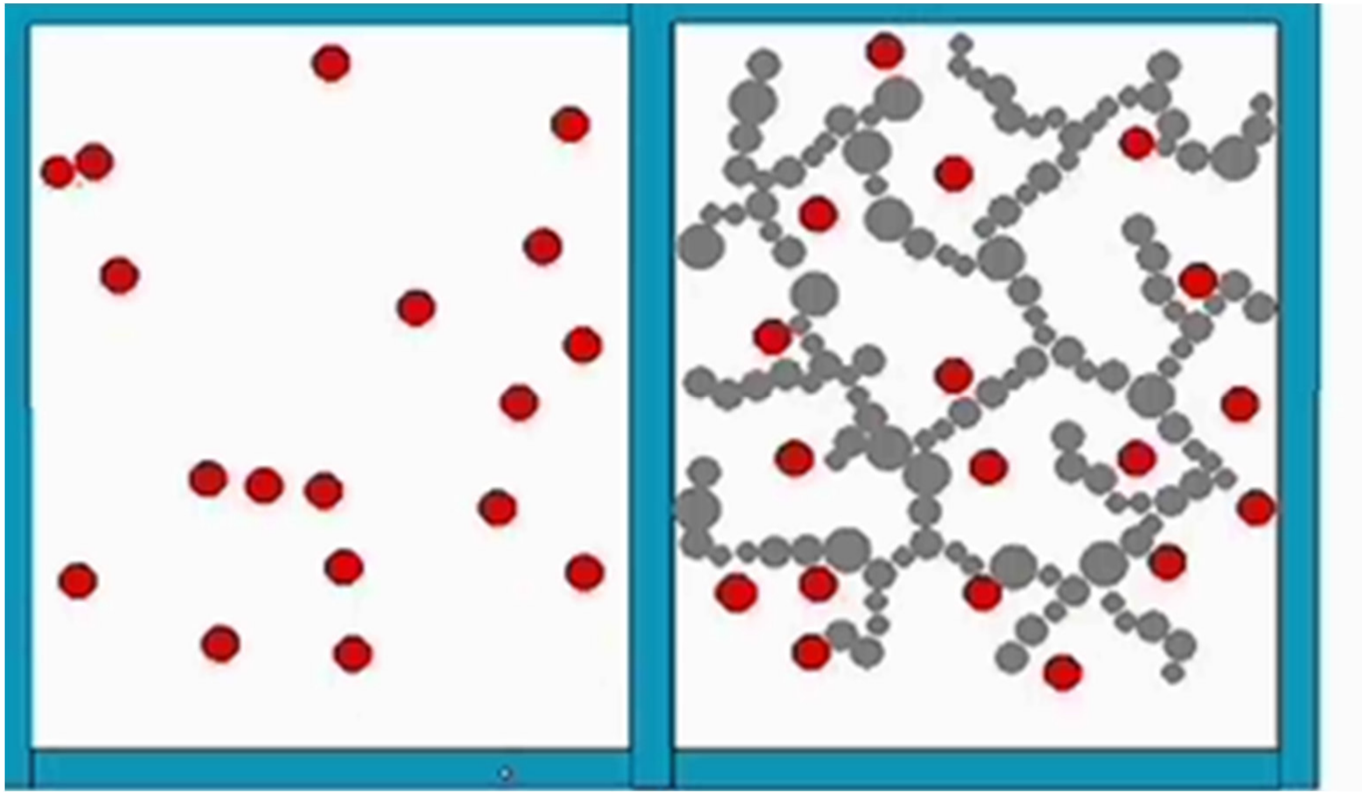


http://www.cabot-corp.com/Aerogel/Building-Insulation/What-Is-Aerogel

Thermal conductivity of Aerogels vs other insulating materials



k aerogel < k air: is it possible?



Pore size in aerogel (20 nm) is lower than free mean path for air molecules: heat transmission less effective in aerogels than in air (convection)

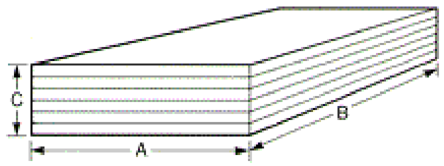
High-k ceramics

- Atoms with similar mass
- High E (diamond E=700-1200GPa)
- Low defects
- Simple crystalline cell (e.g. cubic)

Diamond	2000 ($W \cdot m^{-1} \cdot K^{-1}$)
Graphite	*anisotropic* → *see next slide*
BN (cubic)	1300
BeO	300
AlN	200
SiC	130

Graphite: anisotropy

Pyrolytic Graphite (PG) is a unique form of graphite manufactured by decomposition of a hydrocarbon gas at very high temperature in a vacuum furnace. The result is an ultra-pure product which is near theoretical density and extremely anisotropic. This anisotropy results from the layered structure as indicated in Figure 1. As an example, PG exhibits a thermal conductivity consistent with the best conductors in the "AB" plane and lower than alumina brick in the "C" direction. Mechanical thermal, and electrical properties are generally far superior to conventional graphites. Typical properties are listed in Table 1.



PG is available as plate or machined shapes and as an impermeable coating on graphite and other substrates.

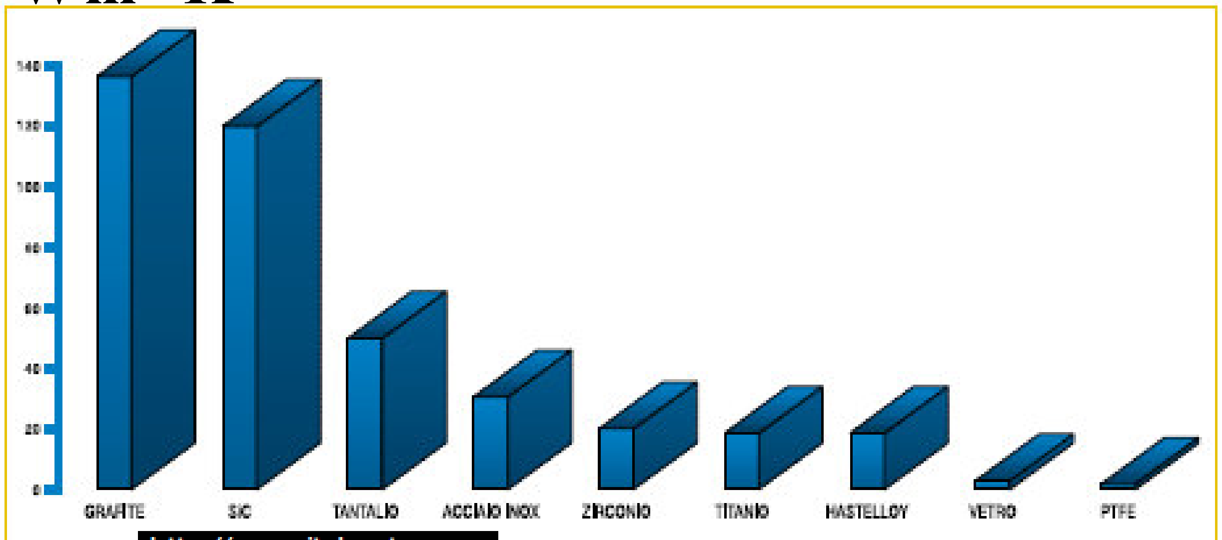
TYPICAL PROPERTIES (RT)*

PROPERTY		VALUE
Density		2.18 - 2.22 g/cc
Flexural Strength	(a b)	18,000 psi (120M Pa)
Tensile Strength	(a b)	12,000psi (80 MPa)
Compressive Strength	(a b)	15,000 psi (100 M Pa)
Young's Modulus	(a b)	3×10^6 psi (20,000 M Pa)

COEFFICIENTS		
Thermal Expansion	(a b)	0.5×10^{-6} cm/cm/°C
	(c)	6.5×10^{-6} cm/cm/°C
Thermal Conductivity	(a b)	400 Watts/Meter °C
	(c)	3.5 Watts/Meter °C
Electrical Resistance	(a b)	0.5×10^{-3} ohm-cm
	(c)	0.5 ohm-cm
Crystal Structure		Hexagonal
(C/2 Spacing)		(3.42Å)
Melting Point (Atmosphere)		Sublimes at 3650 °C

Applications of high k ceramics: SiC heat exchangers

$W \cdot m^{-1} \cdot K^{-1}$



http://www.italprotec.com

Thermal conductivity in ceramics: the influence of grain size

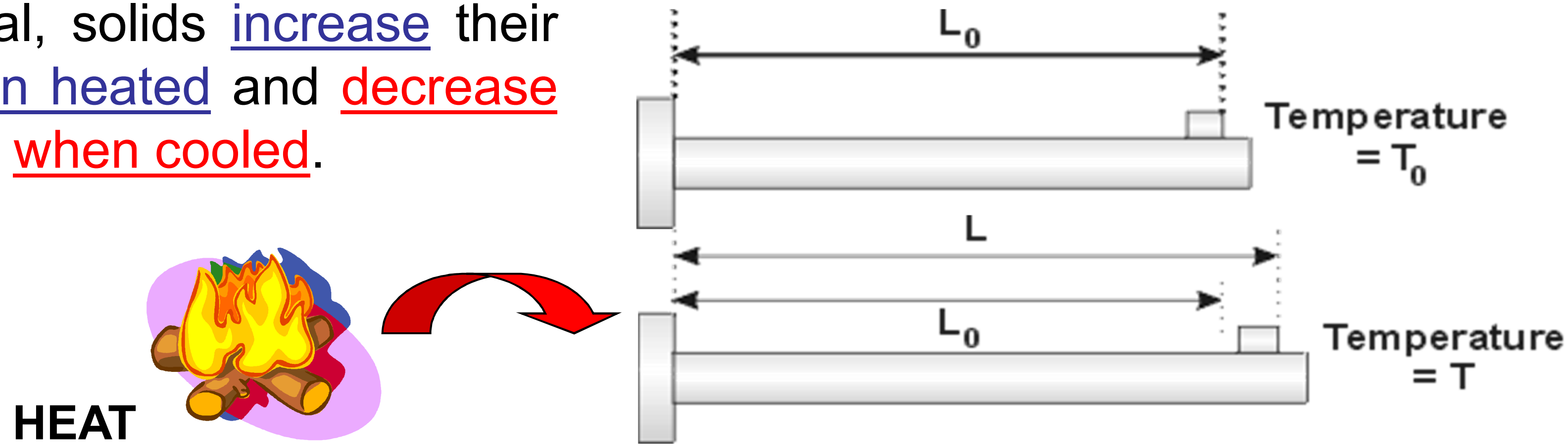
- Effect of grain size:
  like in the case of mechanical resistance (strength), the increase of thermal resistance, k^{-1}, is related to the grain size d, being inversely proportional to $\sqrt{d}$:

$$k^{-1} \propto d^{-1/2}$$

- For ceramics to have high thermal conductivity, **MICROSTRUCTURES with LARGE GRAINS** are useful to minimize the effect of phononoic deviation by grain boundaries and the effect of phases/impurity at grain boundaries.

Thermal expansion

In general, solids increase their size when heated and decrease their size when cooled.



The rod is at a temperature $\mathbf{T_0}$ and has a length of $\mathbf{L_0}$.

We heat the rod up to a temperature $\mathbf{T} > \mathbf{T_0}$, and its length increases to $\mathbf{L}$

We shall call the change in temperature:

$$\Delta \mathbf{T} \equiv \mathbf{T} - \mathbf{T_0}$$

and the corresponding change in length:

$$\Delta \mathbf{L} \equiv \mathbf{L} - \mathbf{L_0}$$

$$\boxed{\Delta L \simeq \alpha L_0 \Delta T}$$

Due to temperature increase ΔT, solids undergo thermal deformation that can be defined as:

$$\varepsilon_t = \frac{l - l_0}{l_0} = \alpha(T - T_0) = \frac{\Delta l}{l_0} = \alpha \Delta T$$

α is called **THERMAL EXPANSION COEFFICIENT** [K^{-1} or $°C^{-1}$]: it indicates how a material expands upon heating

Thermal expansion is reflected by an increase in the average distance between the atoms with respect to their equilibrium positions

⇩

the increase of volume due to thermal expansion is due to the increase of interatomic bond length

Heating ⇨ Increase in thermal vibrations ⇨ Increase in the length of chemical bondings ⇨ **THERMAL EXPANSION**

Material	α (10^{-6}/K) at room T
• **Polymers**	
Polypropylene	145-180
Polyethylene	106-198
Polystyrene	90-150
Teflon	126-216
• **Metals**	
Aluminum	23.6
Steel	12
Tungsten	4.5
Gold	14.2
• **Ceramics**	
Magnesia (MgO)	13.5
Alumina (Al_2O_3)	7.6
Soda-lime glass	9
Silica (cryst. SiO_2)	0.4

increasing α

- Thermal expansion coefficient
  - SiC = 4.3×10^{-6} /K (20 - 1000°C)
  - Al_2O_3 = 8.6×10^{-6} /K (0 - 1000°C)
  - Si_3N_4 = 3.2×10^{-6} /K (25 - 1225°C)
  - Comparison with metals
    - Cu = ~18×10^{-6} /K (20 - 300°C)
    - Al = ~26×10^{-6} /K (20 - 300°C)
    - carbon steel = ~13×10^{-6} /K (20 - 400°C)

Metals	$\alpha \approx$ 5-25 10^{-6} K^{-1}
Ceramics	$\alpha \approx$ 0.5-15 10^{-6} K^{-1}
Polimers	$\alpha \approx$ 50-400 10^{-6} K^{-1}

$\alpha = 10\times10^{-6}$ C^{-1} means that, starting from about room temperature, at 1000 °C elongation is 1%

Note: actually, α Is not constant but (slightly) varies with T

CTE and bond strength

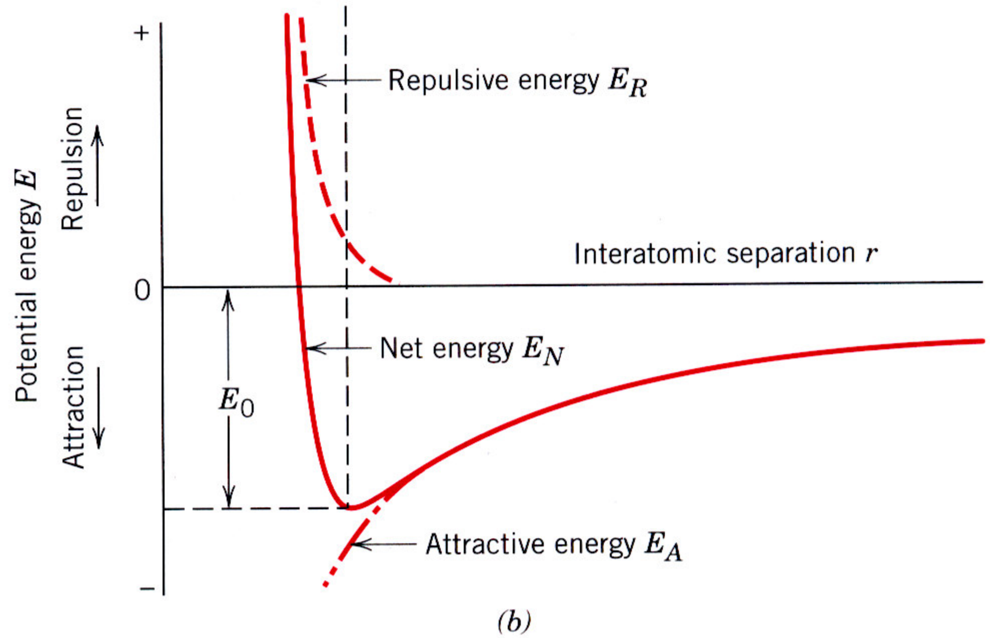
Strongly bonded solids,

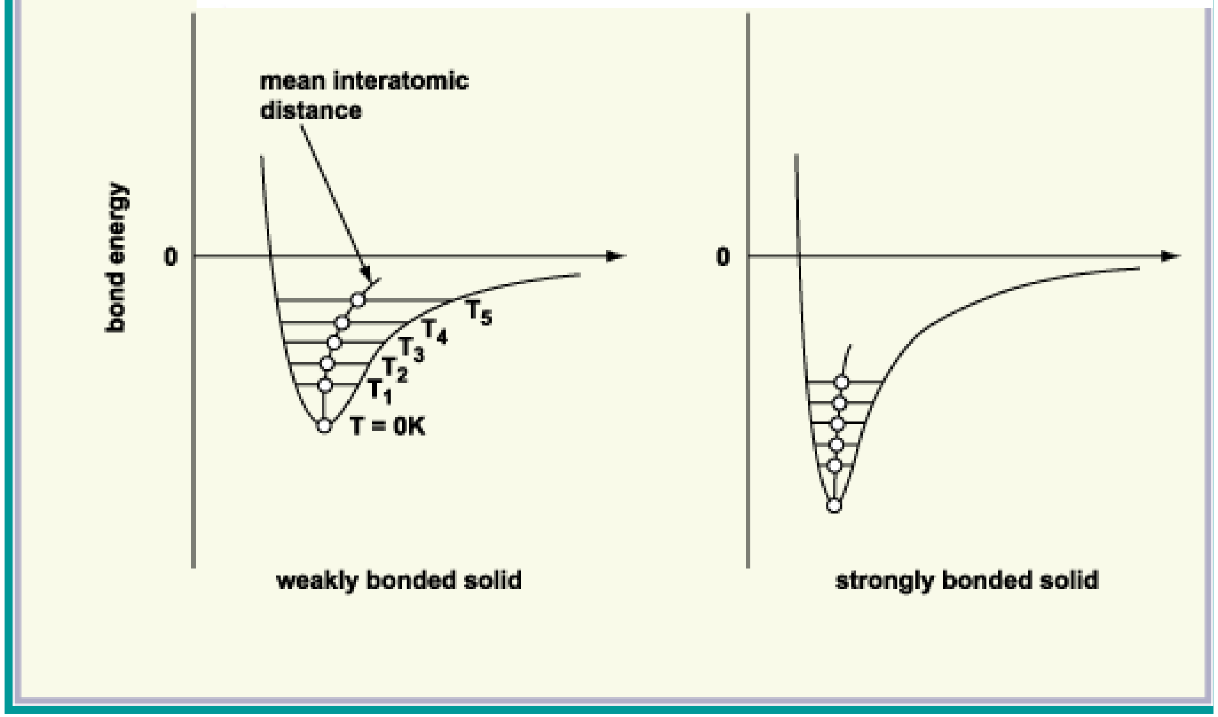
high E,

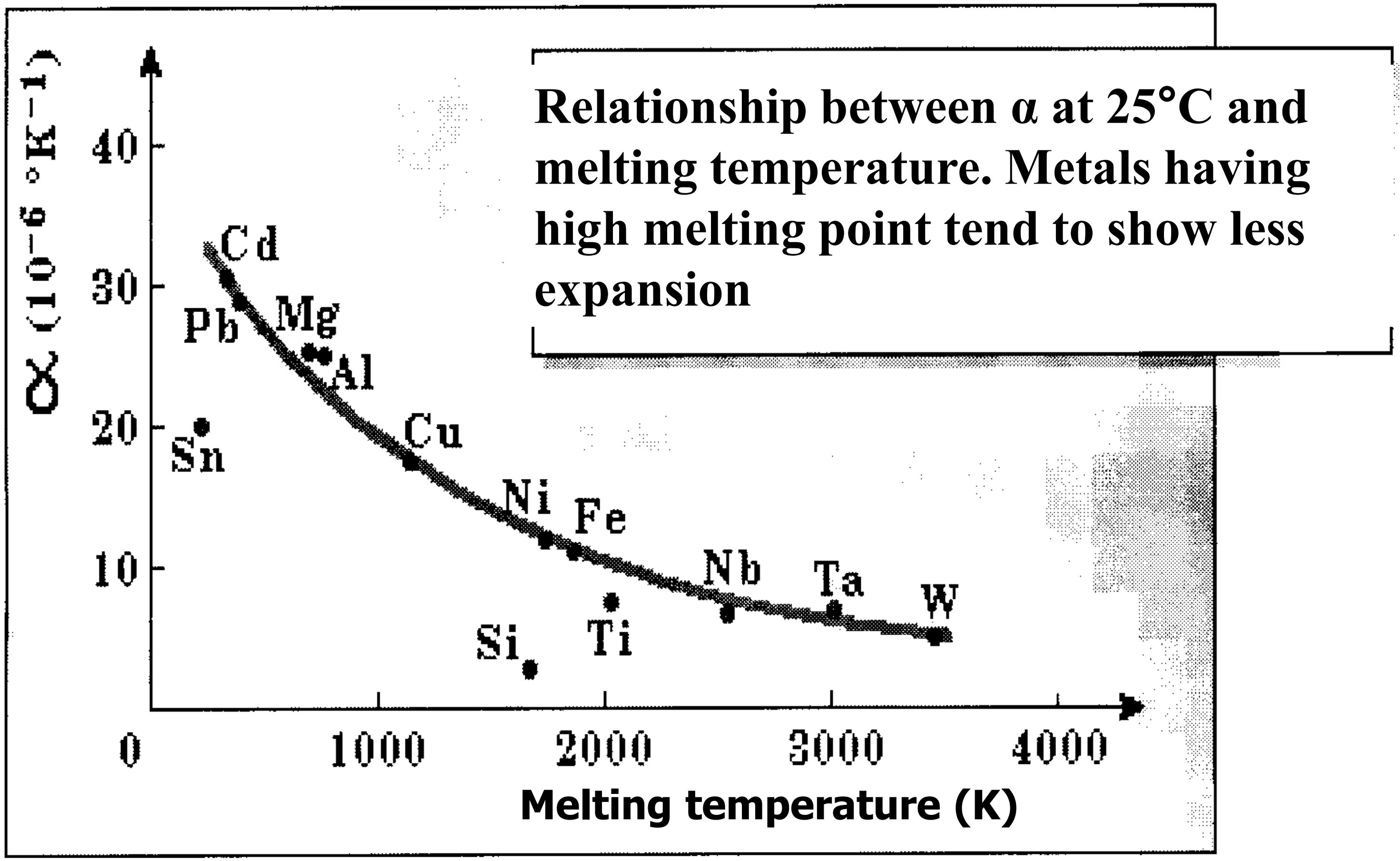
high melt T

⇩

Lower α

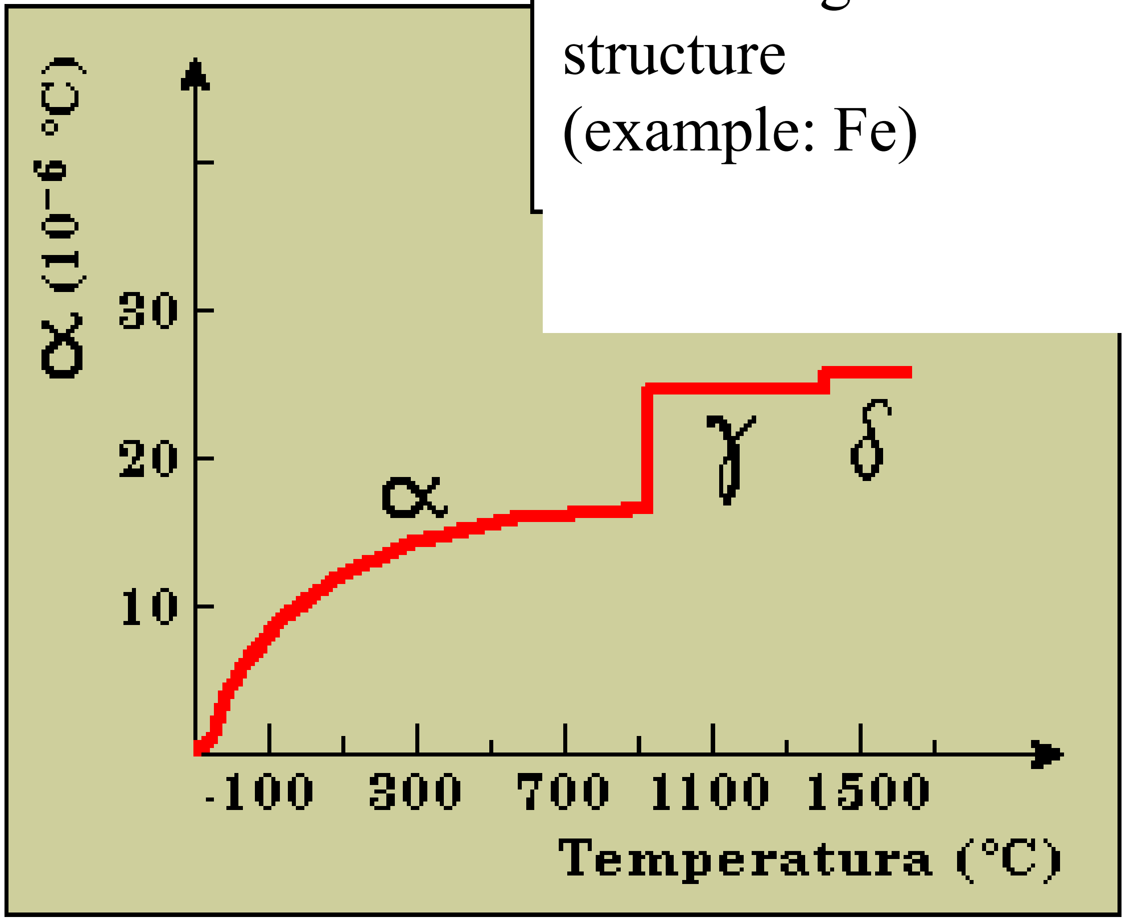






Strong chemical bond: **high elastic modulus**, **high melting point**, **low thermal expansion coefficient**

CTE changes with the crystalline structure
(example: Fe)



Importance of α in materials choice

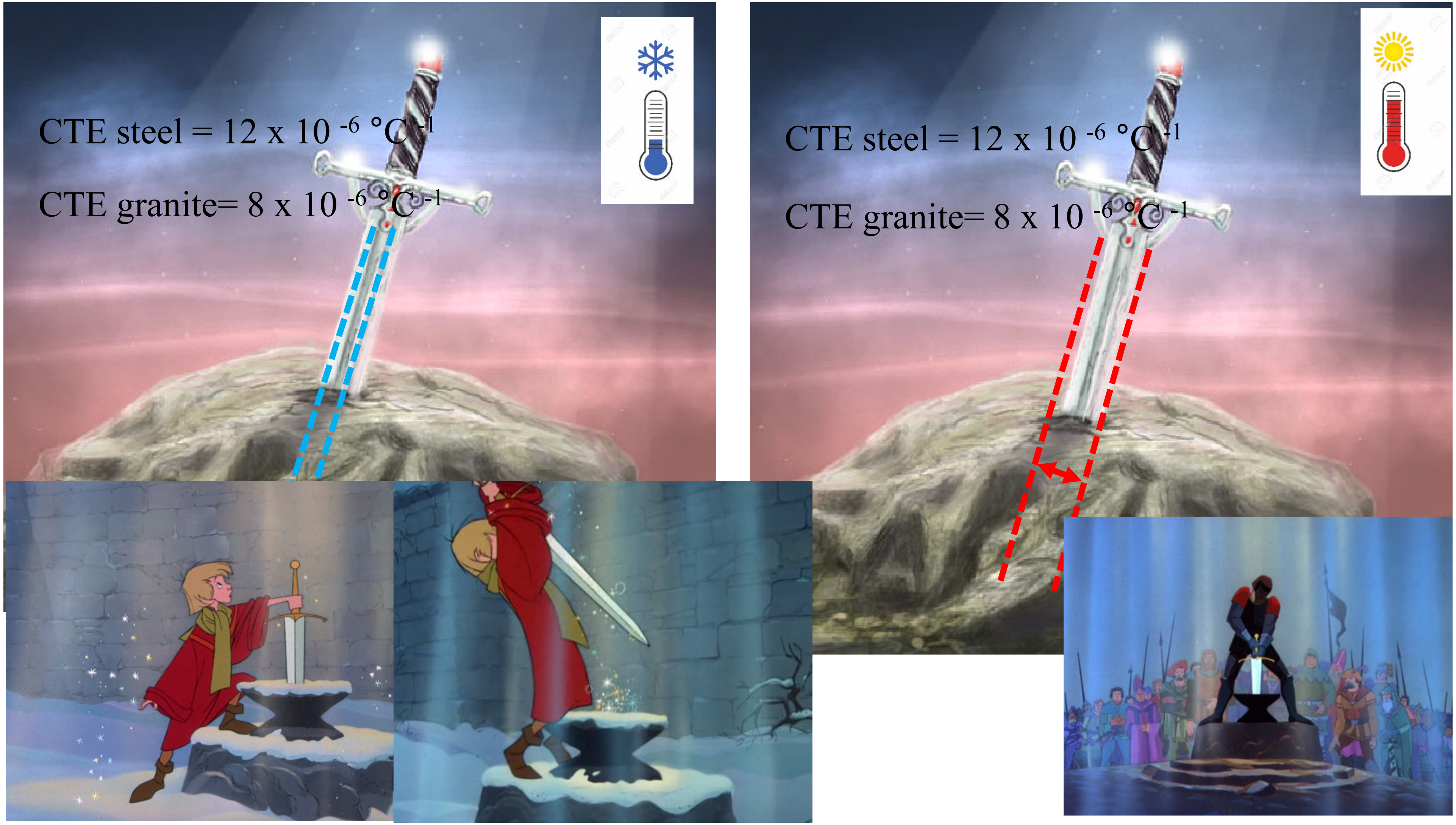
Material design and choice should take into account the **thermal expansion of materials that are put in contact**

Careful attention should be devoted to materials in components working at high temperature, used as a coating or subjected to sudden temperature changes → α **mismatch should be reduced** and appropriate "voids" might be foreseen inside components in order to allow thermal expansion

Examples:

- to manufacture a **metallic joint using polymer resins** as joining material, additives (glass fibers or ceramics) should be put in the polymer with the aim of reducing its thermal expansion coefficient

- Fe/steel reinforcement in concrete: coupling between concrete matrix ("artificial ceramic") and metal-


CTE steel = 12 x 10 -6 °C -1
CTE granite= 8 x 10 -6 °C -1
CTE steel = 12 x 10 -6 °C -1
CTE granite= 8 x 10 -6 °C -1

Thermally induced stresses

$$\frac{(l_t - l_0)}{l_0} = \alpha(T_t - T_0)$$

- Ductile materials
- Brittle materials

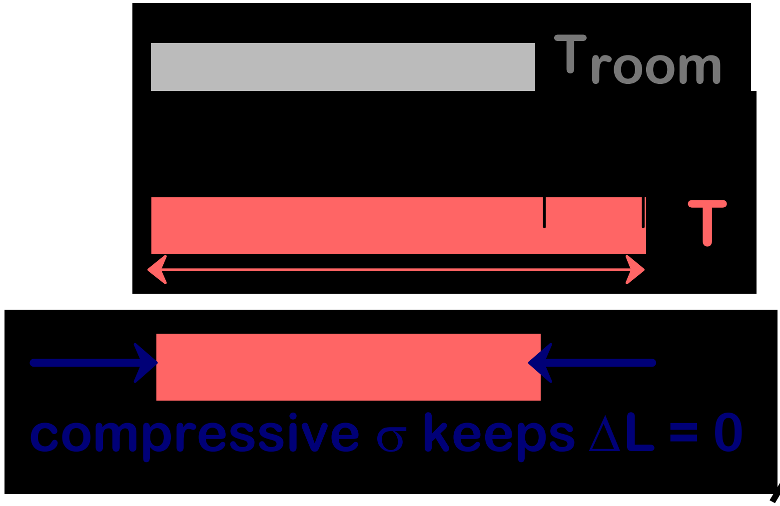
$$\frac{\Delta l}{l_0} = \alpha \Delta T$$

$$\sigma = E\,\varepsilon = E\,\Delta l / l_o = E\,\alpha\,\Delta T$$

Thermally induced stresses: example

- Brass cylinder (E= 100 GPa, σ_y= 172 MPa, α = **20 x 10^{-6} /K**), no thermal expansion allowed, which is the maximum T it can reach without plastic deformation? (room T= 20 °C)

$$\sigma = E\,\varepsilon = E\,\Delta l / l_o = E\,\alpha\,\Delta T$$



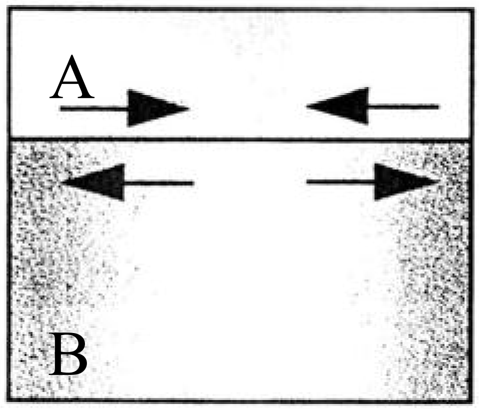
100 GPa **20 x 10^{-6} /K**

$$\sigma = E(-\varepsilon_{thermal}) = -E\alpha(T - T_{room})$$

172 MPa **T= 106°C** **20°C**

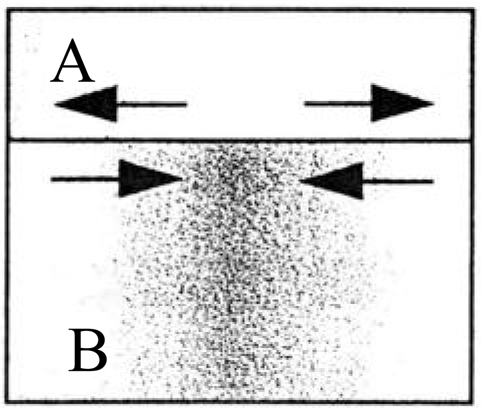
Joined materials with different thermal expansion coefficients- Residual stresses during cooling

- If $\alpha_B > \alpha_A$, during cooling to room T, material A is in compression mode
- If $\alpha_B < \alpha_A$, during cooling to room T, material A is in tensile mode



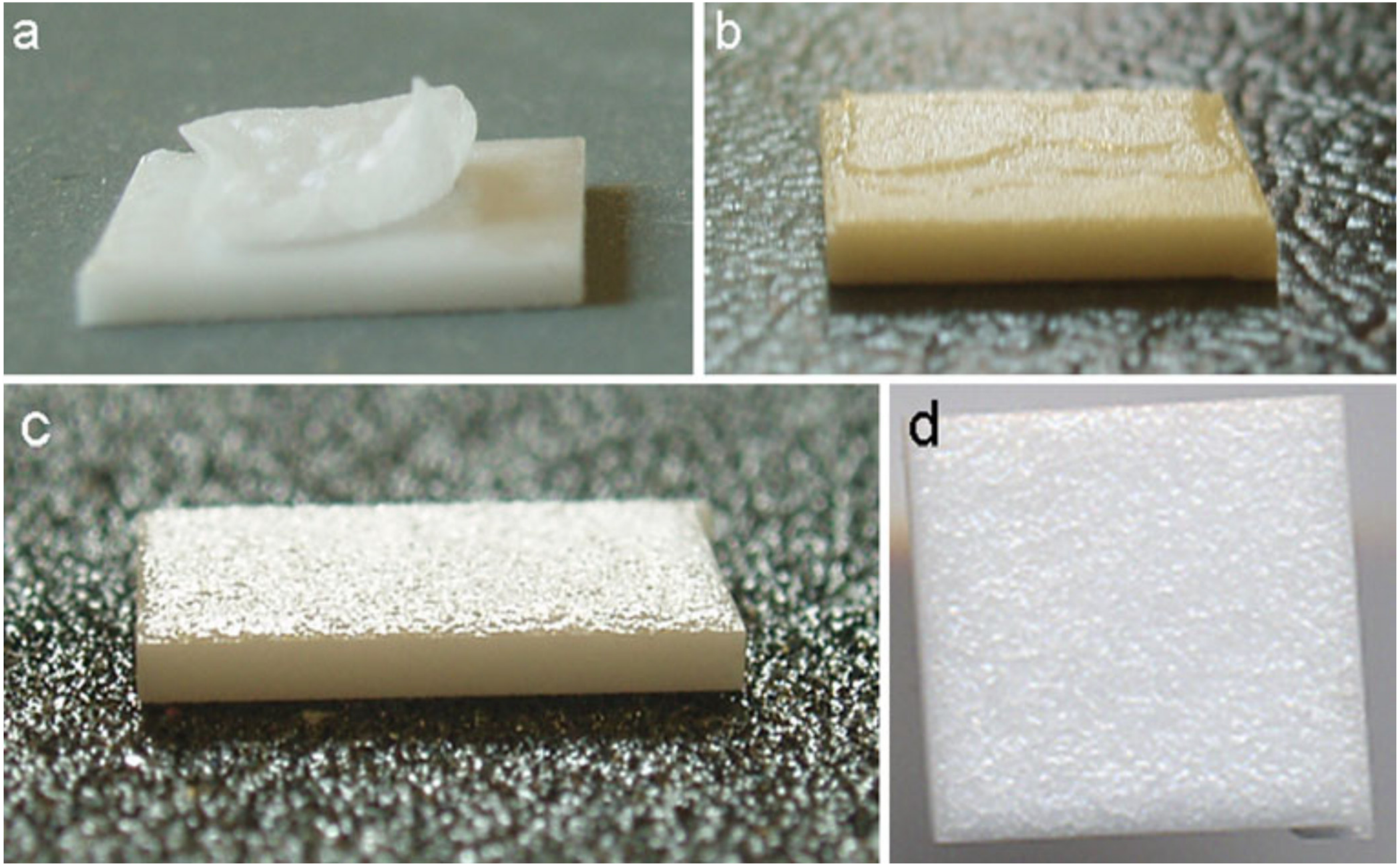
RT

$\alpha_B > \alpha_A$



RT

$\alpha_B < \alpha_A$


a
b
c
d

Stress at the interface between two solids having different CTE

$$\sigma = \frac{E_M E_C}{E_M + E_C}(T_b - T_a)(\alpha_M - \alpha_C)$$

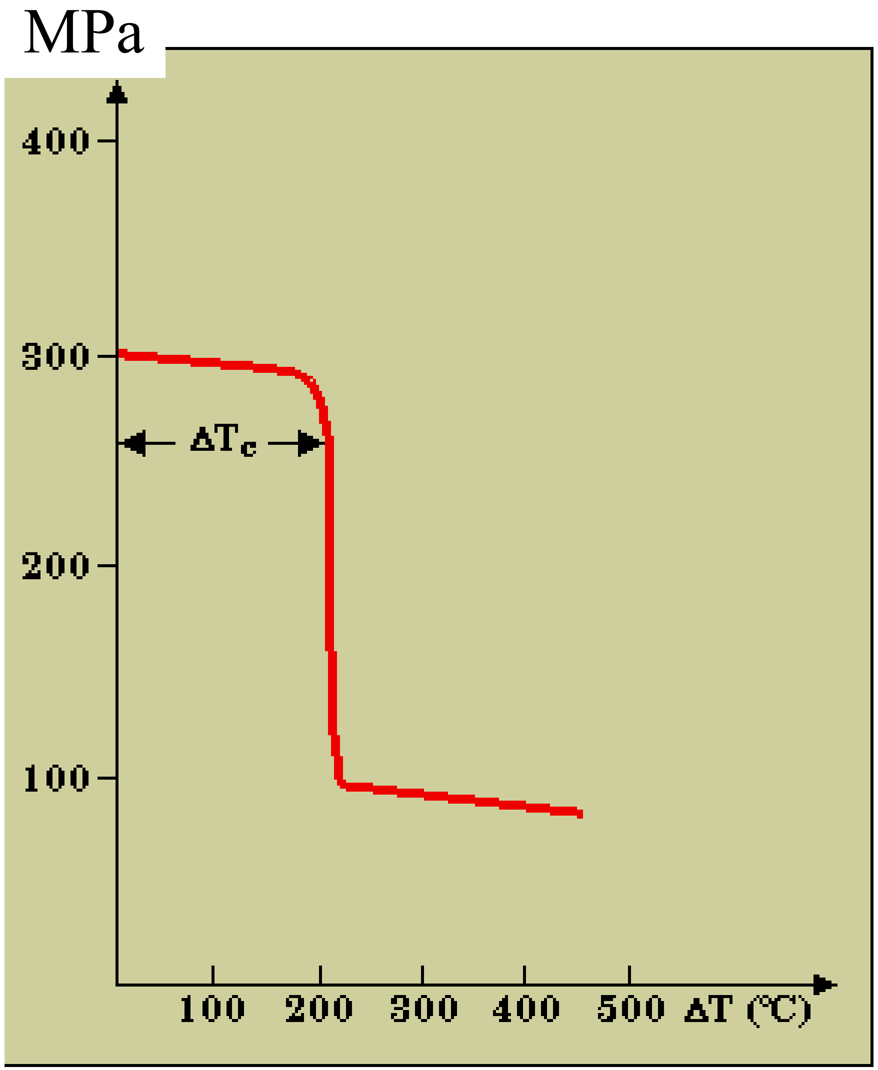
σ = Stress at the interface between two solids having different CTE

E = Young modulus (M=metal, C=ceramic)
T = temperature (Tb –Ta = ΔT)
α = CTE (M=metal, C=ceramic)

Thermal shock resistance (TSR)

- First approximation of temperature variation (ΔT) without failure for a given material

$$TSR \cong \frac{\sigma_f k}{E\alpha}$$



- flexural strength, σ_f
- coefficient of thermal conductivity, k
- elastic modulus, E
- TEC, α

BEST (ideal) MATERIAL: high k and low α

(but it does not exist!)

	Materiale	C_p *(J/kg-K)*[a]	α_ℓ $[(°C)^{-1} \times 10^{-6}]$[b]	k *(W/m-K)*[c]
			Metalli	
Al	Alluminio	900	23.6	247
Cu	Rame	386	17.0	398
Au	Oro	128	14.2	315
Fe	Ferro	448	11.8	80
Ni	Nichel	443	13.3	90
Ag	Argento	235	19.7	428
W	Tungsteno	138	4.5	178
Steel	Acciaio 1025	486	12.0	51.9
Stainless steel	Acciaio inossidabile 316	502	16.0	15.9
brass	Ottone (70Cu-30Zn)	375	20.0	120
	Kovar (54Fe-29Ni-17Co)	460	5.1	17
	Invar (64Fe-36Ni)	500	1.6	10
	Super Invar (63Fe-32Ni-5Co)	500	0.72	10
			Ceramici	
	Allumina (Al_2O_3)	775	7.6	39
	Magnesia (MgO)	940	13.5[d]	37.7
	Spinello ($MgAl_2O_4$)	790	7.6[d]	15.0[e]
Silica glass	Silice fusa (SiO_2)	740	0.4	1.4
Soda-lime glass	Vetro calce-sodico	840	9.0	1.7
Boro-silicate glass	Vetro borosilicato (Pyrex™)	850	3.3	1.4
			Polimeri	
	Polietilene (alta densità)	1850	106–198	0.46–0.50
	Polipropilene	1925	145–180	0.12
	Polistirene	1170	90–150	0.13
	Politetrafluoroetilene (Teflon™)	1050	126–216	0.25
	Fenolo-formaldeide, fenoliche	1590–1760	122	0.15
	Nylon 6.6	1670	144	0.24
	Poliisoprene	—	220	0.14

DUCTILE MATERIALS

(e.g. metals, polymers)

Thermal stresses can induce **plastic deformation** (unavoidable)

BRITTLE MATERIALS

(e.g. ceramics, glasses)

Thermal stresses often lead to **fracture**

avoidable!

Modification of material properties (**mainly** α) by:

- changing chemical composition (e.g. adding other oxides in glasses;
- fabrication of composites (second phases with different TEC)
- pay attention to the presence of pores (crack "stopper", but **k and mechanical strength** $\downarrow$)

SUMMARY

THERMAL PROPERTY	MEANING	CORRELATION TO MATERIAL CHARACTERISTICS
THERMAL CAPACITY	Material ability to absorb heat from the external surroundings	Depends on atomic weight
THERMAL CONDUCTIVITY	Material ability to transfer thermal energy from a system at high temperature to another system at lower temperature	Related to the presence/absence of free electrons and to the propagation of phonons
THERMAL EXPANSION	Material attitude to increase its dimensions upon heating	Depends on the strength of the chemical bonds
THERMAL SHOCK RESISTANCE	Material resistance to repeated, high thermal variations (stresses)	Depends on thermal expansion coefficient and thermal conductivity